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The new Y Chromosome Haplotype Reference Database

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Highlights

- Free online Y Chromosome STR Reference Database for use in forensic casework
- Worldwide repository of over 136,000 9-locus, over 84,000 17-locus, over 25,000 23-locus haplotypes and 15,000 Y SNP profiles from 917 sampling locations in 128 countries
- Search using Excel sheets or GeneMapper exports instead of typing
- Assessment of haplotype frequencies (observed and estimated)
- Enhanced interpretation tools like LR-based mixture calculation

Abstract: After opening the first version of an internet-accessible worldwide reference database of Y chromosome profiles 14 years ago and six years after the last major relaunch the new YHRD 4.0 repository and website has been rolled-out. By November 2014 about 136k 9-locus haplotypes, among these 84k 17-locus haplotypes, 25k 23-locus haplotypes and 15k Y SNP profiles from 917 sampling locations in 128 countries have been submitted by more than 250 institutes and laboratories. In geographic terms, about 39% of the YHRD samples are from Europe, 32% from Asia, 16% from South America, 6% from North America, 4% from Africa and 2% from Oceania/Australia. Worldwide collaboration is the driving force for the rapid growth of the database and this, in turn, allows the evaluation and implementation of enhanced interpretation tools (variable

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frequency estimators, LR-based mixture and kinship analysis, Y-SNP-based ancestry assessment).

Keywords: Database, Y Chromosome, Haplotype, Y-STR, Frequency Estimation, Metapopulation

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Introduction

Y chromosome STR profiling is the most advanced method to analyse the male proportion of female/male cell mixtures. In addition a combination of Y-STRs and Y-SNPs provides valuable investigative leads on the geographical ancestry of the male trace donor. The Y Chromosome Haplotype Reference Database (YHRD) assists the interpretation of profiles consisting of such markers [1]. The analytical concept requires high-resolution and sensitive chemistry as well as large reference sample collections where all major ancestral founder lineages of world populations are represented in appropriate numbers. Interpretation and calculation of match probabilities then follows a genealogical approach, where haplotype frequencies are reported in groups of spatially distributed populations (Metapopulations) sharing a common ancestry and thus a similar pool of deep-rooting lineages. To describe the hierarchy of metapopulations we use a knowledge-based terminology, which incorporates linguistic and geographical resources (e.g. Eurasian-European-Western European (instead of "Caucasian")). The further development of a terminology that could replace a frequently used but diffuse or misleading historical vocabulary as e.g. "Hispanics" or "Caucasian" (in US) or "Mestizo" (in Latin America) is a major challenge but without alternative for Y chromosomal profiles which often (inherently) carry detailed information about geographical or linguistic ancestry.

The amount of studies on Y-STR haplotypes distributions has increased enormously in the past years with no signs of slowing down. Studies cover most regions and even remote niches of the global landscape (Figure 1). In the last two years a new generation of highly discriminative Y-STR kits with more than 20 Y-STRs (PowerPlex® Y23 System, Promega, Madison, USA and Yfiler® Plus, Life Technologies, Foster City, USA) arrived which have sparked an avalanche of new studies. For example, within only 10 months after the release of the Powerplex Y23 kit nearly 20k 23-locus haplotypes have been collected from 129 populations in 51 countries [2]. Nearly all these single-centre and multi-centre studies set-up by forensic institutions are submitted to the YHRD prior to publication in peer-reviewed forensic or human genetics journals. The sheer amount of empirical material of the highest forensic quality allows testing of new mathematical models to calculate the weight of evidence of a Y-STR haplotype match by means of likelihood principles. The proposed approaches with variable estimators (surveying, coalescence, discrete Laplace) mainly concern the frequency estimation of rare haplotypes, whereas frequencies of common haplotypes can be readily estimated using counts and Clopper-Pearson confidence intervals (augmented counting, kappa inflation) [3-7,23]. Those estimation methods, which generate results within reasonable computing time, are now implemented in the YHRD. To keep pace with the rapid developments in all three areas, i.e. data generation, chemistry and mathematics, the YHRD was completely remodeled to become more operational in everyday forensic practice. This paper describes the basic principles and the most important changes to previous versions of the database.

1. Structure and Basic Principles

1.1 Technical Set-Up

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YHRD has been developed following the behavior driven development (BDD) principle [8] for the frontend and using the test driven development (TDD) principle [9] for the database backend. This ensures that YHRD will work as reliable and trustworthy as possible by defining test cases and test scenarios that are tested automatically with each new release.
The web interface itself is written in Ruby [10] using the popular Ruby-On-Rails framework [11]. All haplotype data is managed by a proprietary In-Memory-Database written in C99 and X86_64-Assembler for fastest possible binary pattern matching [12]. Statistical tools are implemented in R [13] based on the excellent clustering library "snow" by Tierney et al. [14] to allow for parallel processing.
Of course, since the objective of the database is to assess haplotype frequencies and not to identify personal matches, we do not store any IDs of the original submissions.

1.2 Input files

As new kits are including increasing number of Y-STR loci, it became inevitable to provide a fully electronic submission facility. To fulfill this need, we implemented a flexible Excel-, XML-, CSV-file upload mechanism. Moreover, users are able to use their GeneMapper (LifeTechnologies Inc., Foster City, USA) export files to search the YHRD. Both should avoid errors introduced by manual data transfer. Of course, manual input is possible as well.

1.3 Datasets

The YHRD is a database, which consists of 5 separate datasets with different resolutions (see Table 1). Datasets are collections of haplotypes all defined for a certain set of Y-STR markers (see Table 1 and Table 2) and might therefore be part of other, bigger datasets. The sets available at YHRD are currently fixed and may change only on new versions. Each result at YHRD (search, tools etc.) is based on one particular dataset, which means that all values are relative to the size and composition of that dataset.

1.4 Metapopulations

The term "metapopulation" was first defined by Levins [15] saying that "a natural population occupying any considerable area will be made up of a number of … local populations". Hanski and Gilpin [16] refer to the term metapopulation to describe spatially distributed population groups, which are interconnected by geneflow and migration. We adapted this terminology to an assemblage of Y-STR profiles spread over a territory, which descend from one or several common ancestral lineages. Population genetic analyses on different Y chromosomal marker sets show that metapopulations are stabilized over time by cultural (e.g. a common language or marital patterns) or geographical factors. Several studies examined the range expansion of Y chromosomal lineages and (sub-) population effects in more depth than we do here [17-21]. The most comprehensive study so far on YHRD datasets [2] shows that with increasing number of Y-STRs included in a marker set, the genetic distances between meta-populations decreased monotonically but are not erased. The distances calculated for 7 up to 23 STRs remain significant, both on a global as well as on a continental level. The YHRD currently includes 33 metapopulations (see Figure 2), which are subject to changes if new samples arrive or variance analysis (e.g. by AMOVA, see section 3.1) confirms divergence of population groups (for an example see Figure S1)

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1.5 National Databases

National databases were established to meet the demands for a local ("national" in political terms) Y-STR reference database as required by legislation. They consist of all population samples collected within the political boundary of the concerned country. The YHRD currently includes 128 national databases with an average size of 1,063 haplotypes (σ = 1,112) extending from about 10 Haplotypes (Solomon Islands with 8 minimal haplotypes and Democratic Republic of the Congo with 13 minimal haplotypes), to about 7,500 (Poland with 7,478 minimal haplotypes, Brazil with 7,506 minimal haplotypes, Germany with 7,675 minimal haplotypes and United States with 7,793 minimal haplotypes), to almost 18,000 (China with 17,792 minimal haplotypes).

1.6 Y-SNPs and haplogroup definition

The extension of Y-STR data with Y-SNP data creates a valuable possibility of clustering haplotypes and to refine existing structuring methods based on non-genetic factors. We include Y-SNP information as sets of Y-SNP marker states (ancestral/derived) and transcode them to haplogroups according to the phylogenetic "minimal Y tree" [22]. The YHRD currently includes more than 15,000 chromosomes typed both for STR and SNP markers, which can therefore assigned to haplogroups.

2 Search the Database

The search result view is made up of panes that are collecting all available values for a certain feature (e.g. the "Worldwide" feature pane collecting all information on the whole dataset). There are two additional feature panes that are giving values based on the appropriate sub-database of the chosen dataset ("National Database" and "Metapopulation"). Each pane consists of two categories of values given:

Observed Actual observed frequency and corresponding confidence interval (CI)[1]

Expected Estimated haplotype frequencies and if applicable the corresponding confidence interval (CI)[1]. There are currently three methods implemented (see section 2.1 for details): Augmented count (n+1/N+1) [3], discrete Laplace [7] and kappa method [4]. Not all approaches necessarily are present at each pane (e.g. kappa method being absent when there are matches observed)

2.1 Frequency Estimation Methods

Together with the actual observed haplotype frequency, an augmented count n+1/N+1 (frequency when adding the haplotype in question to the dataset) is given.

2.1.1 Discrete Laplace

[1] Confidence intervals are calculated using Clopper-Pearson [23].

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The discrete Laplace distribution approximates properties of the Fisher–Wright model of evolution and can be used to estimate Y-STR haplotype frequencies by modeling the frequency as a relation of the composition of the haplotype in question compared to each of the pre-calculated subpopulation centers [7]. This estimation method is only available for adequately sized metapopulations and datasets. When estimating the worldwide haplotype frequency, the average over all metapopulation estimations is given. A disproportionately high contribution of one metapopulation to the overall frequency estimator is stated explicitly.

2.1.2 Kappa Method

Derived from Ewens' work [24] Brenner's kappa (κ) method estimates the haplotype frequency by correcting the simple counting method using κ, with κ being the proportion of singletons (haplotypes that occur only once) in the dataset [4]. Simulation studies [6] showed that the kappa method provides useful frequency information in cases of rare haplotypes. It is therefore taken as a frequency estimator in cases when there is no haplotype match observed.

2.2 Ancestry Information

When adding the "Ancestry Information" feature to a result sheet for any targeted haplotype the incremental minimal haplotype dataset (9 loci) which is present in all sampled individuals is searched. Three different views on the distribution of matching minimal haplotypes provide data, which helps the user to assess the phylogenetic ancestry and the biogeographical origin of the chromosome:

Metapopulation frequency distribution: A graphical representation of the relative frequency of the minimal haplotype for each metapopulation is given. A bolder font indicates a significantly higher relative frequency of the minimal haplotype in question.

Y-SNPs: The composition of associated Y-SNPs to the matched minimal haplotypes is given as a pie chart. All haplotypes without Y-SNP information will be mapped to the phylogenetic root "Y".

National database frequencies: For each match of the minimal haplotype the relative as well as the absolute frequency in the appropriate matching national database is given as a bar plot.

Heat Map: A spatial projection of relative minimal haplotype frequency is given as an interactive world map, with each dot representing a individual sampling site. Non-matches are colored in blue whereas red dots are those with at least one minimal haplotype match. An underlying heat map gradient in red helps spotting sampling sites with high relative frequencies.

3 Associated Tools

3.1 AMOVA / MDS

The AMOVA (pairwise F_{ST}/R_{ST} calculation) [25] has been carried over from the last version of the database. Beginning with the submission of a data file (Excel, CSV or the like)

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composed of one or many population sample(s), the researcher selects any population sample or whole national databases of the appropriate YHRD dataset for comparison. Note that we decided to limit the number of haplotypes involved in a calculation to 10,000 following the principle of fair use.
All selected single population samples and/or national databases are used to calculate a matrix of pairwise F_{ST}/R_{ST} together with their corresponding *p*-values (based on 10,000 iterations of random rearrangement). Based on that matrix the first Principal Components of a two-dimensional Principal Component Analysis (PCA) is evaluated and taken as the starting configuration for a Sammon Mapping [26], a nonlinear metric multidimensional scaling method, which is then plotted as an exchangeable PDF file.
The selected algorithm will cluster population samples / national databases based on limits given by the researcher (e.g. based on an F_{ST}/R_{ST} threshold).
Figure S1 gives an example of an AMOVA analysis of 52 population samples from China representing 17,278 individuals in the minimal haplotype dataset.

3.2 Mixture analysis

We implemented the LR-based Y-STR mixture approach of Wolf et al. [27] using a recursion in the number of unknown contributors and the observed haplotype frequencies given by the YHRD. The mixture tool will calculate the likelihood ratio (LR) of the donorship versus the non-donorship of the suspect to the trace taking into account the given known contributors (if any) and the number of additional unknown contributors (if any). The tool will accept manual input and electronic files (see section 1.2).

3.3 Kinship analysis

YHRD provides a relatively simple Kinship Index (KI) calculation for two different scenarios: father-son and two-brothers. Both are using the well-known formulas [28,29] in case of mutation and non-mutation with the extension that the frequencies of unobserved suspect and/or known contributor haplotypes are estimated by their augmented count. The tool will accept manual input and electronic files (see section 1.2).

4 Contribution and Updates

The YHRD will be updated about four times a year. Submissions using standardized file formats are received, evaluated and accepted by the YHRD, after passing the internal quality assessment. A unique accession number is issued for each submitted population sample.

Required sample sizes and procedures of quality assessment are published in the updated ISFG guidelines for the publication of population data [30]. Accepted studies with accession number are published both in the YHRD and in a peer-reviewed forensic journal. At each result page the YHRD release (date and version) and accession date is noted. This information needs to be included in any reports.

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5 Availability

Published datasets can be retrieved upon request from the YHRD, provided that an outline of the research project is submitted for publication on the website.

6 Safety, Security and Validation

High-resolution Y-STR profiling with the current generation of Y-STR kits may approach the identification level in some cases. To protect the privacy of submitted data, we decided to encrypt each communication with the YHRD using Transport Layer Security (TLS). Additionally the whole system is built for anonymous usage meaning that we don't use any external analytics or logging facility and users don't have to register or login to use the YHRD at all.

To fulfill the requirements of a court-going forensic expert system we defined almost 230 individual test cases covering every algorithm at the YHRD. Those tests will ensure the software is working within expected specifications. For example a random Y-STR database is created (with known frequencies) and each result of a random set of searches will be checked against their expectations. If any of those tests fail, the release/update will not be published to the public. After successful review of the failure, the release/update will be published to the public and all dataset specific release information will be updated automatically.
To keep track of possible incompatibilities and/or bugs, we run a bug tracking system (https://redmine.yhrd.org) along with the YHRD. This ensures that a failed or bogus behavior of the YHRD will be immediately turned into an electronic test case and tested therefore with each release/update as described above.

As each report should be documented, we place the date of the action and the date and release of the YHRD at each final result page.

Summary

The Y Chromosome haplotype Reference database (YHRD) is a computing platform, which allows acquisition, distribution, evaluation and interpretation of forensic DNA datasets. Dedicated solely to Y chromosome polymorphisms (Y-STRs, Y-SNPs) YHRD experiences and features can serve to help other online resources in the rapidly evolving field of forensic genetics. New technologies (e.g. massive-parallel sequencing results) and new mathematical tools can be integrated and linked to the data sets. Since datasets grow and tools evolve, such platforms need to be updated regularly. With the current version of the YHRD we have facilitated the process of registration, accelerate the allocation of data and reduced the response time. To achieve this we have reduced the output of a typical haplotype search. The default results page presents the frequency estimates (constant and variable estimators) for the selected dataset, without loading tables of matching and neighboring haplotypes, maps and numbers in subpopulations. However, additional features can or must be added, namely frequency estimates in national or metapopulation databases or contextual ancestry information. All details on the datasets (e.g. contributors) or the markers (e.g. mutation rates) can be retrieved via the link "Resources". Likewise, tools for data or match interpretation can be approached by a link where calculations for typical Y-STR applications (e.g. mixture

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interpretation or kinship analysis) can be performed. The reduction of output and the concentration on the most important features (frequency estimators) make the new YHRD much more operational in daily forensic practice than previous versions. It remains nevertheless a fully annotated database where all documents on the data resources and the validation processes for submitters are accessible. This is one important prerequisite for court use of such repositories. In view of the wide application of this database we seriously take our responsibility to provide education. The website has screened tutorials, a manual, a FAQ section and lists of all relevant publications. The biannual Y-User Workshop, established in 1996 with 9 conferences in 5 countries so far, becomes the most important forum for scientific discussion on the forensic application of Y chromosome markers which eventually led to the establishment of the new YHRD [31].

URL

Y Chromosome Haplotype Reference Database: https://yhrd.org

Acknowledgement

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Captions of Tables and Figures

Table 1: Compositions of the five YHRD datasets as of release 48 (November 2014). Each row represents a separate dataset with a name and the defining markers followed by the count of haplotypes and the number of populations, national databases and metapopulations structuring the dataset.

Table 2: Distribution of identical haplotypes in the five YHRD datasets as of release 48 (November 2014). Each row represents the amount of haplotypes being present in the appropriate dataset exactly *n* times. E.g. *n=1* will give the count (and relative percentage) of all singletons included in each corresponding dataset.

Figure 1: Worldwide map of national databases at the YHRD.

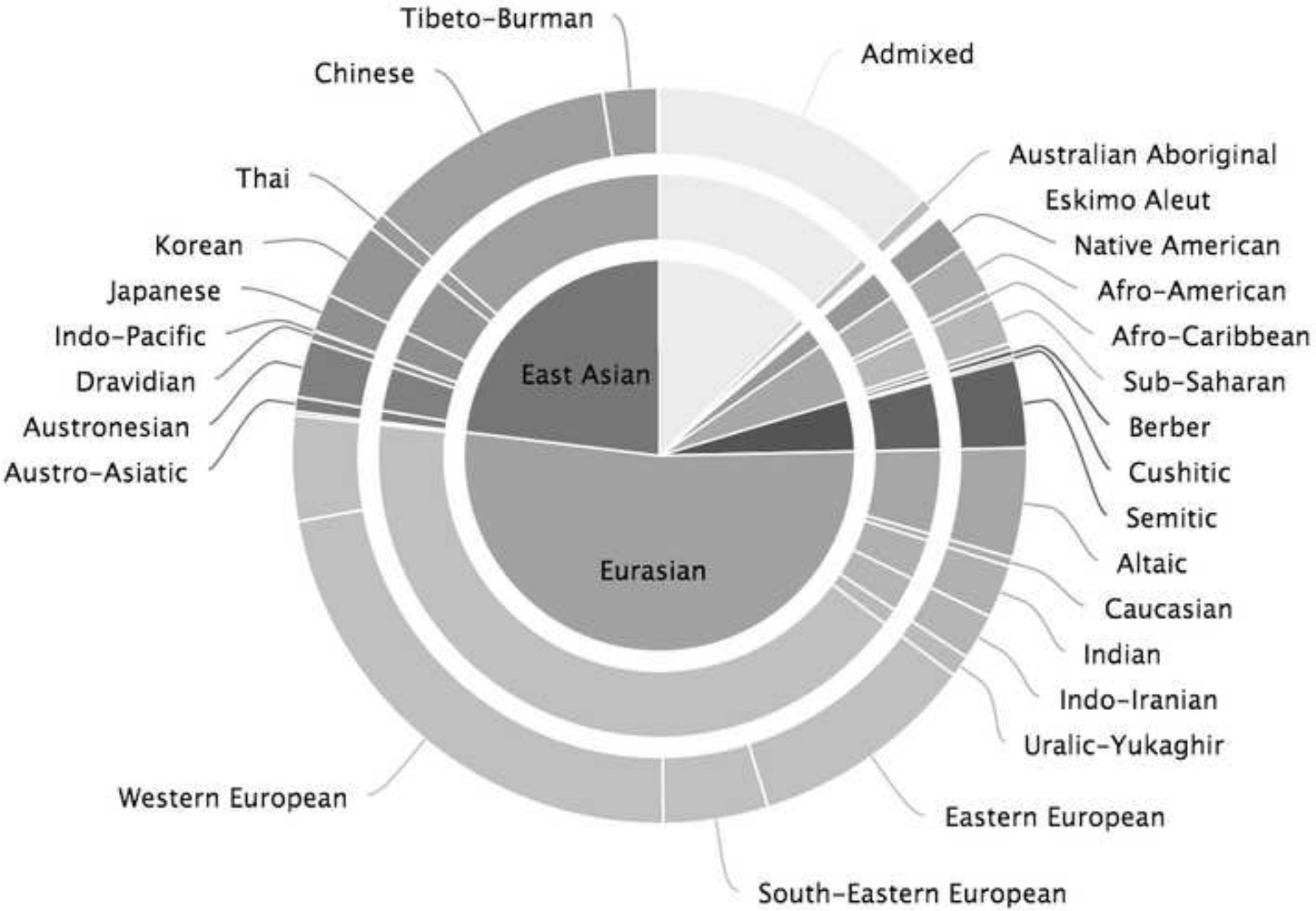
Figure 2: Structure and terminology of metapopulations defined for the YHRD. Note that the second layer is not labeled to improve readability.

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Figure 1



Figure 2


Tibeto-Burman
Chinese
Thai
Korean
Japanese
Indo-Pacific
Dravidian
Austronesian
Austro-Asiatic
East Asian
Admixed
Australian Aboriginal
Eskimo Aleut
Native American
Afro-American
Afro-Caribbean
Sub-Saharan
Berber
Cushitic
Semitic
Altaic
Caucasian
Indian
Indo-Iranian
Uralic-Yukaghir
Eurasian
Western European
Eastern European
South-Eastern European

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Table 1

Dataset	YSTR markers	Number of Haplotypes	Number of Population Samples	Number of National Databases	Number of Metapopulations
Minimal	DYS19, DYS389I, DYS389II, DYS390, DYS391, DYS392, DYS393, DYS385	136,184	917	128	33
Promega PowerPlex Y12	Minimal + DYS437, DYS438, DYS439	96,735	659	115	31
Applied Biosystems Yfiler	Promega PowerPlex Y12 + DYS448, DYS456, DYS458, DYS635, YGATAH4	84,256	572	107	31
Promega PowerPlex Y23	Applied Biosystems Yfiler + DYS481, DYS533, DYS549, DYS570, DYS576, DYS643	25,118	175	54	28
Applied Biosystems Yfiler Plus	Applied Biosystems Yfiler + DYS576, DYS627, DYS460, DYS518, DYS570, DYS449, DYS481, DYF387S1, DYS533	1,297	13	8	7

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Table 2

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n	Minimal	Promega PowerPlex Y12	Applied Biosystems Yfiler	Promega PowerPlex Y23	Applied Biosystems Yfiler Plus
1	25,215 (~19%)	33,510 (~35%)	55,946 (~66%)	22,685 (~90%)	1,171 (~90%)
2	5,869 (~4%)	5,860 (~6%)	5,761 (~7%)	914 (~4%)	22 (~2%)
3	2,537 (~2%)	2,223 (~2%)	1,534 (~2%)	100 (<1%)	2 (<1%)
4	1,347 (~1%)	1,125 (~1%)	660 (~1%)	38 (<1%)	2 (<1%)
5	891 (~1%)	612 (~1%)	312 (<1%)	8 (<1%)	4 (<1%)
6	615 (<1%)	412 (<1%)	197 (<1%)	3 (<1%)	1 (<1%)
7	415 (<1%)	272 (<1%)	122 (<1%)	2 (<1%)	1 (<1%)
8	319 (<1%)	217 (<1%)	81 (<1%)	3 (<1%)	0
9	269 (<1%)	171 (<1%)	51 (<1%)	0	0
10	226 (<1%)	125 (<1%)	46 (<1%)	1 (<1%)	0
>10	1,714 (~1%)	934 (~1%)	229 (<1%)	4 (<1%)	3 (<1%)

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Figure 1 in Color

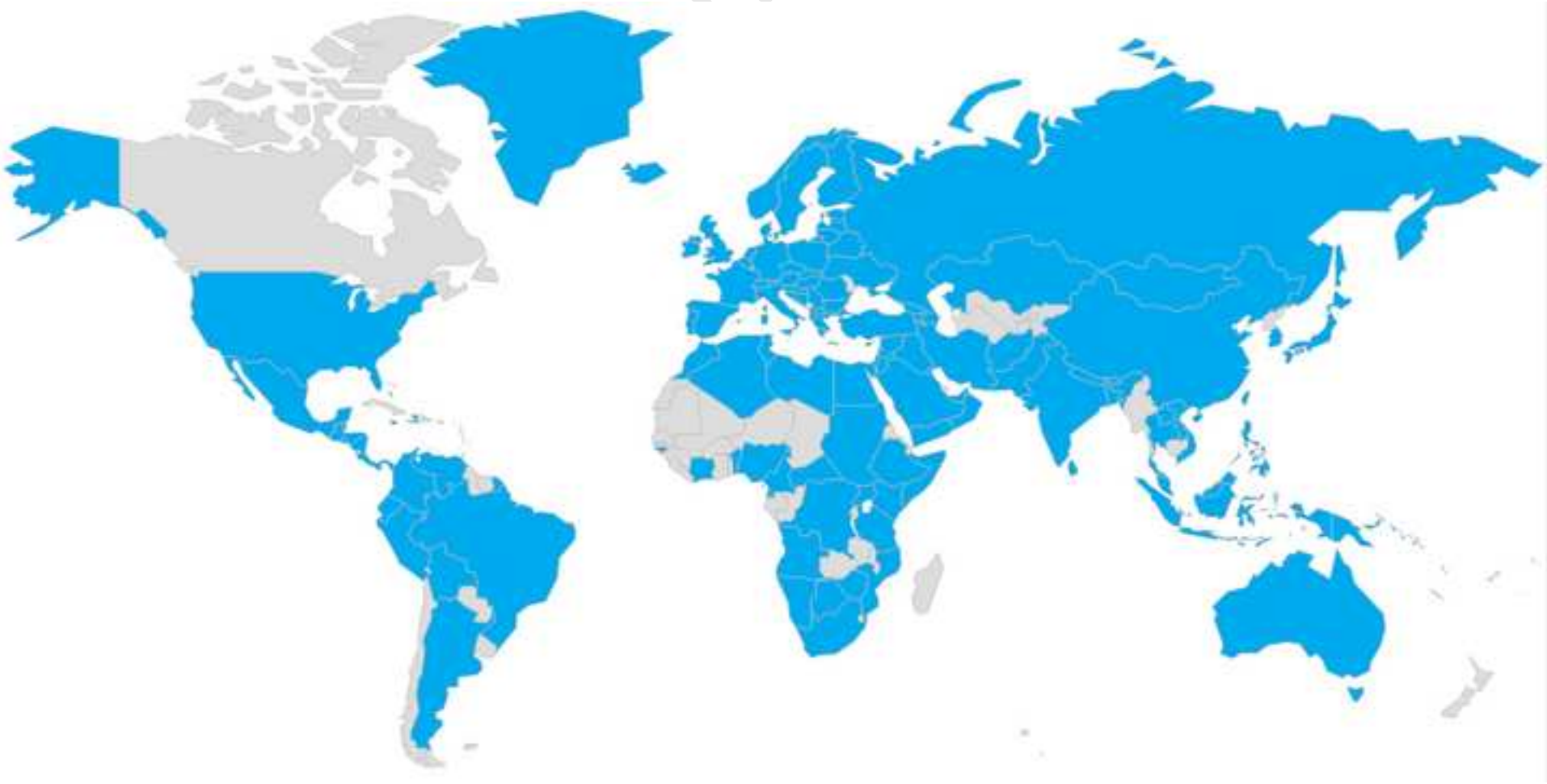


Figure 2 in Color

